Abhay Kapoor

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EDUCATION

Wellesley High School Wellesley, MA
Sophomore *2025 – Present*

- **Overall GPAs: Weighted: 4.75/5.00; Unweighted: 3.75/4.00**
- **AP/Honors Courses:** AP Computer Science Principles, AP Modern World History, Physics 9H, Chemistry 10H, Pre-Calculus Honors
- **Honors, Awards, and Publications:** State Team Varsity Tennis Player for Wellesley High School

Courses taken Outside of School

- **Math:** AOPS Scholars Math 11: Calculus, MIT Linear Algebra Lecture Series
- **Computer Science:** Harvard CS50 and CS50AI, Inspirit AI Bootcamp

RESEARCH AND PERSONAL PROJECTS

Neural Network from Scratch Wellesley, MA
Personal Project *Jul 2026 – Present*

- Built a NumPy tensor library from scratch and an automatic differentiation engine, including support for broadcasting and gradient accumulation, to understand the mathematical foundations of deep learning frameworks like PyTorch
- Developed core parts of neural networks: linear layers, activation functions, forward propagation, and backprop
- Designed a beginner-friendly machine learning API that abstracts matrix operations while allowing users to build, train, and evaluate neural networks using simple Python lists, making machine learning more accessible to beginners
- Previous projects can be found at my Git Hub page for examples, such as my Connect 4 AI using self-play RL

3D Printing and CAD Projects Wellesley, MA
Personal Project *2024 – Present*

- Utilized Onshape resources provided by school to create own models for 3D printing, including a working table vise
- Modeled parts for school Science Olympiad and Robotics team; worked with others to deploy parts in a real setting

LEADERSHIP AND INVOLVEMENT

Vertex Math Project Wellesley, MA
Co-Founder, Volunteer Tutor/Teacher, Curriculum Coordinator *Aug 2026 – Present*

- Designed multiple lessons ranging from virtual high school competition math workshops for middle school students to engaging seminars in a local library for 60 elementary school students, 15 students per seminar for four seminars
- Teaching 100 kids around the Boston area as well as 17 different states united under the Vertex Math community
- Expanding project to more employees and future math competitions that mock actual test kids need to prepare for

Newton-Wellesley Hospital Volunteering Newton, MA
Volunteer in Day Surgery Recovery room *Sep 2026 – Present*

- Assisted in discharging patients, cleaning rooms, stocking, making voiding trial kits, and other services required
- Collaborated with nurses and staff to support patient care, respond to unit needs, and maintain a nice environment

Tennis Player and Coach Wellesley, MA
Varsity Tennis Player for Wellesley School Team *2023 – Present*

- Currently top 7 player on the Wellesley School Team, and last year was part of the state team as a freshman
- Teach tennis at local courts to beginners, hoping to inspire them the same way the game inspired me
- Looking to be a future captain of the Wellesley tennis team and continue to spend time with friends over the game

Science Olympiad Wellesley, MA
Member of Main Team *2023 – Present*

- Working on events: Electric Vehicle, Experimental Design, and Engineering CAD for competition at Dartmouth U
- A few things I have done: created autonomous cars using C++, speed-modeled engineering drawings in a 3D modeling software (Onshape), designed an experiment using given materials and a topic that it should follow

SKILLS AND INTERESTS

Skills – Python, PyTorch, CAD, Tutoring/Teaching

Interests – Artificial Intelligence, Competition Math, 3D Printing, Watching Tennis, Running, Rainbow Six Siege